Development of cerebral sulci and gyri in fetuses of cynomolgus monkeys (Macaca fascicularis). II. Gross observation of the medial surface.

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This study aimed to clarify chronological sequences of the appearances of sulci and gyri on the medial cerebral surface and its relation to the regional development of the cerebrum in cynomolgus monkeys. The lengths of cingulate and calcarine sulci were measured, and the ratios of these lengths to fronto-occipital length were estimated as indices of the size of the "frontoparietal" and "occipital" regions, respectively. The relative length of cingulate sulcus showed a biphasic increase: a slow phase from EDs 100 to 110, and a rapid phase from EDs 110 to 130. The gyri in the "frontoparietal region" were convoluted in the limbic cortex during the initial slow phase and in the neocortical region during the rapid phase. The relative length of calcarine sulcus lineally increased between EDs 90 and 130, and the gyri in the "occipital region" generated in a dorso-ventral manner: the gyrus convolutions occurred first in the "phylogenetically older" striate and dorsal extrastriate cortices, and then in the "phylogenetically newer" ventral extrastriate cortex. The results suggest that the chronological order of appearance of sulci and gyri is closely associated with the order of phylogenetical development of the cerebral cortex. The present study provides a standard reference for the development of cerebral sulci and gyri of cynomolgus monkeys together with our previous study (Fukunishi et al. Anat Embryol 211:757-764, 2006).

[Structure of the neocortex of the Baikal ringed seal (Pusa sibirica Gm.)].

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The sulci and gyri of the neocortex, as well as cyto-, synaptoarchitectonics and neuronal composition of the sensomotor (brain area) have been studied in the Baikal ringed seal. The structure of the sulci and gyri have been found to be similar to that in carnivores. The following specific features have been revealed in the brain of this endemic species: a thick layer I, presence of giant pyramidal cells in the layer III, large mitochondria in the presynaptic parts and dendrites. The results obtained are discussed concerning adaptation to semiaqueous way of life and to diving.

Embryonic mouse medial neocortex as a model system for studying the radial glial scaffold in fetal human neocortex.

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Neocortex is the evolutionarily newest region in the brain, and is a structure with diversified size and morphology among mammalian species. Humans have the biggest neocortex compared to the body size, and their neocortex has many foldings, that is, gyri and sulci. Despite the recent methodological advances in in vitro models such as cerebral organoids, mice have been continuously used as a model system for studying human neocortical development because of the accessibility and practicality of in vivo gene manipulation. The commonly studied neocortical region, the lateral neocortex, generally recapitulates the developmental process of the human neocortex, however, there are several important factors missing in the lateral neocortex. First, basal (outer) radial glia (bRG), which are the main cell type providing the radial scaffold to the migrating neurons in the fetal human neocortex, are very few in the

mouse lateral neocortex, thus the radial glial scaffold is different from the fetal human neocortex. Second, as a consequence of the difference in the radial glial scaffold, migrating neurons might exhibit different migratory behavior and thus distribution. To overcome those problems, we propose the mouse medial neocortex, where we have earlier revealed an abundance of bRG similar to the fetal human neocortex, as an alternative model system. We found that similar to the fetal human neocortex, the radial glial scaffold, neuronal migration and neuronal distribution are tangentially scattered in the mouse medial neocortex. Taken together, the embryonic mouse medial neocortex could be a suitable and accessible in vivo model system to study human neocortical development and its pathogenesis.

The appearance of the otter brain.

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On ten otter brains the appearance of the exterior and interior brain structures was studied. Viewed from above there are two forms of brain, a triangular and a longitudinal one. In the triangular form the gyri are sufficiently developed with a considerable number of transient gyri and in the longitudinal form the gyri are wider and separated by considerably deep sulci. The olfactory bulbs, the olfactory tract, the gyrus olfactorius medialis and the gyrus olfactorius lateralis are less pronounced in the triangular form than in the longitudinal one. In the triangular form of the brain all gyri, except for the gyrus sylvius, are vertically positioned. The brain stem is wider in the triangular form and narrower in the longitudinal one. On the medial aspect of the brain, the cingulate gyrus, on which transient gyri are apparent, is well marked. The cingulate gyrus is slightly wider than in the longitudinal form. Below the splenium corporis callosi there is the tuberculum gyri dentati to be found by a shallow sulcus which is separated from the gyrus corporis callosi. By scaling the brain matter from outside, the pes hippocampi is exhibited. In the longitudinal form it is more sloping and separated from the base of the brain by a thin layer of brain substance. In the triangular form of the brain the cerebellum is covered by the posterior portions of the hemisphere and in the longitudinal form projects rather backward. On the frontal sections of the cerebrum the subcortical grey masses are sufficiently developed.

Cerebriform Cutaneous Lesions in Pemphigus Vegetans.

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Pemphigus vegetans is an autoimmune bullous disorder characterized by vegetating lesions commonly over the flexures. A 42-year-old female patient came with pemphigus vegetans presenting with interesting cerebriform morphology of the cutaneous lesions over the flexures. Cerebriform tongue, a morphology with typical pattern of sulci and gyri over dorsum of the tongue is a well-known sign seen in pemphigus vegetans. Interestingly, we noticed the typical sulci and gyri pattern in the skin lesions of pemphigus vegetans over the flexures of the body. This clinical sign can be used as a clue in the diagnosis of pemphigus vegetans. Morphology and physical characteristics are important for the diagnosis of the disease. Clinical signs always give a clue to the probable or possible diagnosis in most of the dermatological conditions.

Sonographic demonstration of the sulci and gyri on the convex surface in normal fetuses using 3D-ICRV rendering technology.

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To demonstrate morphological alteration of the sulci and gyri on the convex surface in normal fetuses using innovative three-dimensional inversion and Crystalvue and Realisticvue (3D-ICRV) rendering technology. 3D fetal brain volumes were collected from low-risk singleton pregnancies between 15+0 and 35+6 gestational weeks. Volumes were acquired from the transthalamic axial plane by transabdominal ultrasonography and were then post-processed with Crystalvue, Realisticvue rendering software and inversion mode. Volume quality was assessed. The anatomic definition of the sulci and gyri was determined according to location and orientation. The morphology alteration and sulcus display rates were recorded in sequential order of gestational weeks. Follow-up data were collected in all cases. 294 of 300 fetuses (294 brain volumes) (98%) with qualified fetal brain volumes were included (n=294, median 27 gestational weeks). 6 fetuses with unsatisfactory 3D-ICRV image quality were excluded. The morphology of the sulci and gyri on the brain convex surface could be demonstrated clearly on 3D-ICRV images. The Sylvian fissure was the first structure to be recognized. From 25 to 30 weeks, other sulci and gyri became visible. An ascending trend in the display rate of the sulci was found in this period. Follow-up showed no detectable anomalies. 3D-ICRV rendering technology is different from traditional 3D ultrasound. It can provide vivid and intuitive prenatal visualization of the sulci and gyri on the brain surface. Moreover, it may offer new ideas for neurodevelopment exploration.

The Cerebral Cortex is Bisectionally Segregated into Two Fundamentally Different Functional Units of Gyri and Sulci.

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The human cerebral cortex is highly folded into diverse gyri and sulci. Accumulating evidences suggest that gyri and sulci exhibit anatomical, morphological, and connectional differences. Inspired by these evidences, we performed a series of experiments to explore the frequency-specific differences between gyral and sulcal neural activities from resting-state and task-based functional magnetic resonance imaging (fMRI) data. Specifically, we designed a convolutional neural network (CNN) based classifier, which can differentiate gyral and sulcal fMRI signals with reasonable accuracies. Further investigations of learned CNN models imply that sulcal fMRI signals are more diverse and more high frequency than gyral signals, suggesting that gyri and sulci truly play different functional roles. These differences are significantly associated with axonal fiber wiring and cortical thickness patterns, suggesting that these differences might be deeply rooted in their structural and cellular underpinnings. Further wavelet entropy analyses demonstrated the validity of CNN-based findings. In general, our collective observations support a new concept that the cerebral cortex is bisectionally segregated into 2 functionally different units of gyri and sulci.

Regularity and variability of functional brain connectivity characteristics between gyri and sulci under naturalistic stimulus.

PMID: 38039888 | Indexed in PubMed

The human cerebral cortex is folded into two fundamentally anatomical units: gyri and sulci. Previous studies have demonstrated the genetical, structural, and functional differences between gyri and sulci, providing a unique perspective for revealing the relationship among brain function, cognition, and behavior. While previous studies mainly focus on the functional differences between gyri and sulci under resting or task-evoked state, such characteristics under naturalistic stimulus (NS) which reflects real-world dynamic environments are largely unknown. To address this question, this study systematically investigates spatio-temporal functional connectivity (FC) characteristics between gyri and sulci under NS using a spatio-temporal graph convolutional network model. Based on the public Human Connectome Project dataset of 174 subjects with four different runs of both movie-watching NS and resting state 7T functional MRI data, we successfully identify unique FC features under NS, which are mainly involved in visual, auditory, emotional and cognitive control, and achieve high discriminative accuracy 93.06 % to resting state. Moreover, gyral regions as well as gyro-gyral connections consistently participate more as functional information exchange hubs than sulcal ones among these networks. This study provides novel insights into the functional brain mechanism under NS and lays a solid foundation for accurately mapping the brain anatomy-function relationship.

Elucidating functional differences between cortical gyri and sulci via sparse representation HCP grayordinate fMRI data.

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The highly convoluted cerebral cortex is characterized by two different topographic structures: convex gyri and concave sulci. Increasing studies have demonstrated that cortical gyri and sulci exhibit different structural connectivity patterns. Inspired by the intrinsic structural differences between gyri and sulci, in this paper, we present a data-driven framework based on sparse representation of fMRI data for functional network inferences, then examine the interactions within and across gyral and sulcal functional networks and finally elucidate possible functional differences using graph theory based properties. We apply the proposed framework to the high-resolution Human Connectome Project (HCP) grayordinate fMRI data. Extensive experimental results on both resting state fMRI data and task-based fMRI data consistently suggested that gyri are more functionally integrated, while sulci are more functionally segregated in the organizational architecture of cerebral cortex, offering novel understanding of the byzantine cerebral cortex.

Ensemble learning for the detection of pli-de-passages in the superior temporal sulcus.

PMID: 36460275 | Indexed in PubMed

The surface of the cerebral cortex is very convoluted, with a large number of folds, the cortical sulci. These folds are extremely variable from one individual to another, and this large variability is a problem for many applications in neuroscience and brain imaging. In particular, sulcal geometry (shape) and sulcal topology (branches, number of pieces) are very variable. "Plis de passages" (PPs) or "annectant gyri" can explain part of the topological variability, namely why sulci have a variable number of pieces across subjects. The concept of PPs was first introduced by Gratiolet (1854) to describe transverse gyri that interconnect both sides of a sulcus, that are frequently buried in the depth of sulci, and that are sometimes apparent on the cortical surface, hence seemingly interrupting the course of sulci and

separating them in several pieces. Nevertheless, the difficulty of identifying PPs and the lack of systematic methods to automatically detect them has limited their use. However, based on a recent characterization of PPs in the superior temporal sulcus, we present here a method to automatically detect PPs in the superior temporal sulcus. Local morphology within the sulcus is characterized using cortical surface profiling, and the three-dimensional PP recognition problem is performed as a two-dimensional image classification problem with class-imbalance. This is solved by using an ensemble support vector machine model (EnsSVM) with a rebalancing strategy. Cross validation and quantitative experimental results on an external dataset show the effectiveness and robustness of our approach.